SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I P.Hemanth Reddy 192365046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ______________

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

① Compare Type-1 and Type-2 Hypervisor for a cloud datacenter Hosting Mission-critical Applications

Introduction

A hypervisor is virtulization software that enables multiple virtual machines (VMs) to run on a single physical server

feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Run directly on hardware	Run on top of a host operating system
Performance	very high	Moderate due to OS overhead.
security	strong isolation and smaller attack	Lower security bcause host OS can be compromised
Resource usage	Efficient utilization	Additional resorce consuption.

Analysis:

Performance

- Type-1 provides near-native performance
- Type-2 experience additional latency because every request passes through the host OS.

Security:

- Type1 - has better isolation and fewer vulnerabilites.
- Type2 - is vulnrable if the host operating system is attacked.

Manageability:

- Type-1 supports clustering snapshots. high availbity and centralized monitoring.
- Type-2 is mainly used for development and testing.

Justification:

For mission-critical cloud application, Type-1 Hypervisor is the best choice because it Provides:

- High performance
- Better security
- High availability
- Effecient resource utilization
- Entepise seclability.

conclusion:

Hypervisors and prefered for production cloud datacenters because they ensure rainlity security and maximum performance. and the perefered prodution are enses data centers and High availity efficient resouce utilization

(2) Workload Isolation Using Virtualization-Architetue

Introduction

virtualization enables multiple organizations to share the same physical harware while keeping workloads isolated.

-How Virtualization Supports Isolation

- Separate virtual Machines for each department
- Hypervisor isolates CPU, memory and storage resources
- Virtual Networks prevent unauthorized communication.
- virtual firwalls enforce security policies.
- storage virtulization separates data volumes
- Access control resiscts administrative privileges

Isolation Exomple

- Finance -VM1
- Healthcare - VM2.
- student portal - VM3

Each VM has:

- Independent operating system
- seperate memory.
- Dedicated storage

Major Attack surfaces

1. Hyparvisor Attacks
   - Hypervisor escape
   - VM breakout
   - pivilege escalation

2. virtual Network Attacks:
   - VM sniffing
   - VLAN hopping
   - ARP spoofing
   - Unarthorized lateral movement

Recommendations

- Use secure Type-1 hyparvisors.
- Enable encryption for storage and network traffic.
- Apply role-based access control
- Regularly update hyparvisor patches

Conclusion:

Virtualization architecture provides stong workloa isolation while reducing harware costs, but hyparvisor and virtual network security must be carefully productecl.

3 VM Host Utilization Analysis.

Given Snapshot

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O latency = 35 ms
- Average VM Boot Time = 170sec

Analysis

CPU Bottleneck

- CPU Usage is extemely high
- VMs complete for processing resources
- Results in slower application response.

Corrective Action.

- Add more CPU cores
- Migrate VMs to another host
- Enable CPU load balancing

Memory Bottleneck

- Memory utilization is clase to maximum.
- Hypervisor may start swapping memory.

Corrective Action

- Increse RAM
- optimize VM memory allocation
- Remove unsed VMs

Storage Bottleneck

- Storage latency of 35 ms is high
- slow disk access delays VM operations

Corrective Action

- upgrade to SSD/NVMe storage
- Use storage caching
- Impove storage RAID configuration

VM Boot Time

- 170 seconds is much higher then expected
- Mainly caused by storage and cpu bottleracks

Corrective Action

- Optimize startup sequence
- Use fast storage
- Reduce unnessary statup service.

Conclusion

The host is overloaded upgarding CPU, RAM and storage along with balancing workloads will improve overall VM performance. Resource contention among VMs.

4 Software Defined Data centers (SDDC) vs Traditional Datacenter.

Introduction

-A software Defined datacenter (SDDC) virtualizes compute, storage, and networking resource allowing centralized software-based management.

Feature	Traditional Datacenters	Software defined data center
compute	physical servers	Virtual machines
storage	Dedicated storage devices	storage vitualization
Network	Hardware switches	software-defined network
Management	Manual	Automated
Scalability	slow	Rapid
Resource provisiong	Manual	on-demand

How SDDC Improves Agility

1. Compute virtualization
   - Dynamic VM creation
   - Live migration
   - Auto scaling

2. storage virtualization
   - storage pooling
   - Thin provisioning
   - Snapshot and replication

Benefits:

- Faster deployment
- Better resource utilization
- Reduced operational cost
- Automated management
- High availabity.
- Disater recuery support

Conclusion

SDDC significanty improves agility by automating compute, storage and network mangement making it more flexible and effient then traditional datacenters.

(5) Multi-cloud federation Identity Problem.

Proble Analysis:

In a milti cloud environment differed cloud providers use diffrent identity systems.
This leads to:

- Multiple user accound
- Authentication failures
- ~~E.~~ Inconsistent permissions
- Difficult user management.
- Increased security risks.

Secure Identity and privacy Design

1. Identity federation
   - Use a centralized Identity provider (IdP)
   - All cloud providers trust the same identity source.

2. Single sign-on (SSO)
   - Users authenticate once
   - Access multiple cloud services secure

3. standard Authentication protocols
- SAML
- OAuth 2.0
- openID connect

4. Multi factor Authentication (MFA)
- Password + OTP or biomatic verification

5. Role-Based Access control (RBAC)
- Assign permissions based on usa roles
- Enforce least privilege.

Benefits.
- Centralized identity management
- Improved security
- Batter privacy
- simplifed administration
- seamless multi-cloud access.

Conclusion:
Implementing Identity federation with SSO MFA RBAC standard authentication protocols ensure and efficient access across multiple cloud Providers while proctecting usa Privacy.